BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: PR – XII (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Min | **Maximum Marks:** 20

Q1. Domain of function f(x)=|x| is____.

(A) (0,∞) (B) (-∞,∞)

(C) (-∞,0) (D) (0,1)

Q2. The function f(x)=√x-3 is continuous for all x and____.

(A) x>3 (B) x≤3

(C) x<3 (D) x≥3

Q3. lim(x→0) $(e^x-1)/x$ = _____.

(A) 0 (B) e

(C) 1 (D) x

Q4. d/dx(cos x) = _____.

(A) sin x (B) -sin x

(C) tan x (D) sec x

Q5. If $f(x)=2x^3-3x^2+4$, then f'(2)=_____.

(A) 10 (B) 12

(C) 14 (D) 8

Q6. lim(x→0) tanx/x = _____.

(A) 0 (B) 2

(C) 1 (D) 3

Q7. If y=sin x then y''' = _____.

(A) sin x (B) -sin x

(C) cos x (D) -cos x

Q8. The graph of a function f(x) is concave downward on (a,b) where:

(A) f''(x)>0 (B) f''(x)<0

(C) f''(x)=0 (D) None of these

Q9. A function f(x) is decreasing if____.

(A) f'(x)>0 (B) f'(x)<0

(C) f'(x)=0 (D) f''(x)=0

Q10. $\int_0^1 x^2 dx$ = _____.

(A) 0 (B) -1/3

(C) 2/3 (D) 1/3

Q11. ∫cotx dx = _____.

(A) ln|cos x|+c (B) ln|sin x|+c

(C) tan x+c (D) ln|tan x|+c

Q12. The slope of the line y-1=0 is_____.

(A) -1 (B) 0

(C) 1 (D) None of these

Q13. The distance between two points A(3,-2) and B(-1,-5) is____.

(A) 10 (B) 11

(C) 5 (D) 4

Q14. The center of circle $x^2+y^2+2gx+2fy+c=0$ is_____.

(A) (g,f) (B) (-g,-f)

(C) (g,-f) (D) (-g,f)

Q15. The parabola $y^2=4px$, if p<0 then it is open_____.

(A) Down (B) Up

(C) Left (D) Right

Q16. If the eccentricity e<1, the conic is called_____.

(A) Ellipse (B) Parabola

(C) Circle (D) Hyperbola

Q17. $y^2/25 - x^2/4 = 1$ is an equation of____.

(A) Circle (B) Parabola

(C) Ellipse (D) Hyperbola

Q18. Order of the differential equation $(d^3y/dx^3)^2 - (d^2y/dx^2)^4 + dy/dx + y = 0$ is_____.

(A) 4 (B) 3

(C) 2 (D) 1

Q19. $f(x,y) = (\sqrt{x} + \sqrt{y})/(x+y)$ is a homogenous function of degree____.

(A) -1/2 (B) 1/2

(C) 0 (D) 1

Q20. $\int_0^1 e^{x^2}\, dx$ has_____.

(A) One actual solution (B) More than one actual solutions

(C) No actual solution (D) Both A and B

SECTION-B

Time Allowed: 2 Hours and 40 minutes

QQ2:. Answer any TEN parts. Each part carries FIVE marks.

1) Find the inverse function of $f(x) = (3x+1)/5$

2) Evaluate $\lim(\theta\to 0)\ (\tan\theta - \sin\theta)/\sin^3\theta$

3) Use first principal rule to determine the derivative of $f(x) = 5x^2 - 3x$?

4) Find dy/dx by using any suitable rule of differentiation $y = x^2e\hat{}(\sin(x))$

5) If $f(x) = \ln(4x+7)$, find f''(x).

6) Evaluate $\lim(t\to 0)\ [\sin(3t)/(3t)\ i + \ln(2+t)\ j]$

7) Find the Maclaurin series expansion for the function $f(x)=\sin^2 x$.

8) Evaluate $\int x^3 \cos x\, dx$

9) Use the integration by parts toe valuate $\int x^2 \cos x\, dx$

10) Find the centroid of the triangle ABC whose vertices are A(3,5), B(4,6) and C(3,-1).

11) Find the equation of parabola with vertex (-3,4) and passes through (5,11) and its line of symmetry is horizontal.

12) If $f(x,y)=x^2y+xy^2$, then find f_x and f_y.

13) Solve the initial value problem $dy/dx = 1/x^2$, $y(2)=0$

SECTION-C

Note: Note: Attempt any THREE questions. All questions carry equal marks.

QQ3:.

(a) Find dy/dx if $y = ((2x-5)/(x-4))^4$

(b) Find the angle θ from the line $2x+4y-10=0$ to the line $5x-3y+1=0$

QQ4:.

(a) Find the equation of tangent to the circle $x^2+y^2=9$ which makes an angle of 60° with x-axis?

(b) Find the critical value of $f(x) = x^2/(x-2)$

QQ5:.

(a) Find the $\int(\pi/2 \text{ to } \pi) \cos(x/2 + \pi)\, dx$

(b) Find an equation of a circle which passes through the points (1,2), (3,-4) and (5,-6).

QQ6:.

(a) Solve the differential equation $x(dy/dx) = x + y$

(b) Find the equation of hyperbola with vertices are at (5,0) and (-5,0) and one focus at (-7,0).